Aeration System

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Oxygenation system



What it is used for

An aeration or oxygenation system supplies air to water in aquaculture or animal-husbandry settings to support dissolved oxygen levels and aquatic animal health. Select the system capacity for the tank or pond, monitor water conditions, and maintain reliable operation, especially during low-oxygen periods.

Main parts

- Air pump, blower, or oxygen supply unit
- Airline tubing and connectors
- Diffusers, air stones, or aeration plates
- Valves and flow regulators
- Electrical supply and protective enclosure
- Dissolved oxygen meter or water-quality monitoring equipment

Before you start

1. Confirm the system type, rated capacity, and suitability for the tank, pond, or holding system.
2. Inspect the pump, tubing, diffusers, valves, electrical cables, and protective covers.
3. Position the pump in a dry, ventilated area protected from water splash and flooding.
4. Check that tubing is securely connected and diffusers are clean and correctly positioned.
5. Verify the electrical protection and earthing according to the installation instructions.
6. Measure dissolved oxygen and check water temperature and animal behaviour before operation.

How to use

1. Ensure the system is installed according to the manufacturer's instructions and local requirements.
2. Open the required valves and confirm that air pathways are clear.
3. Switch on the pump or blower and check that air is flowing through all intended diffusers.
4. Adjust airflow within the rated range to achieve suitable aeration without excessive turbulence or stress to animals.
5. Observe bubble distribution, water circulation, pump noise, and any unusual vibration.
6. Monitor dissolved oxygen and water temperature at suitable intervals, especially during hot weather, at night, or when stocking density is high.
7. If oxygen levels fall or the system fails, follow the site's emergency aeration plan and activate approved backup equipment.

8. Inspect tubing and diffusers for blockages or leaks; isolate power before removing or repairing components.
9. Record operating status, water-quality readings, adjustments, and maintenance actions.
10. At shutdown, follow the site procedure and ensure alternative oxygen support is available if animals depend on continuous aeration.

⚠ Safety

- Only trained personnel should install, adjust, or repair the system.
- Keep electrical equipment dry and use appropriate protective devices and earthing.
- Never handle plugs, switches, or electrical connections with wet hands or while standing in water.
- Do not block air inlets or operate a pump beyond its rated capacity.
- Avoid entering tanks or confined spaces unless the site has an approved entry procedure and appropriate safeguards.
- Maintain backup aeration or oxygen supply where loss of aeration could endanger animals.

Care & storage

- Clean diffusers and air stones regularly to prevent clogging.
- Inspect tubing, valves, seals, and connectors for leaks or deterioration.
- Clean pump air filters and service the blower according to the maintenance schedule.
- Keep the area around the pump dry, clean, and unobstructed.
- Test alarms, backup systems, and dissolved oxygen meters at the required intervals.

Common mistakes to avoid

- Choosing a pump that is too small for the water volume or animal load.
- Ignoring low dissolved oxygen readings or abnormal animal behaviour.
- Allowing diffusers to clog or tubing to leak.
- Performing maintenance on powered equipment without isolation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.